

Limbic Journey

Full Game Design Document

1. Game Overview

Limbic Journey is a survival and memory-management game inspired by how the limbic system processes memory. The player acts as the brain’s memory center, deciding what to remember, what to forget, and how to organize information while guiding a traveler from their home to a distant castle over 15 nights.

2. Core Educational Goals

Real Memory Concept	Game Mechanic
Selective attention	Daily memory limit
Long-term consolidation	Night memory selection
Emotion strengthens memory	Emotional event tagging
Repetition strengthens memory	Memory growth system
Context matters	Relation-based hazards
Retrieval cues	Sound, smell, and visual clues
False memories	Risky memory merges

3. Core Gameplay Loop

Phase	What Happens
Day	Player observes 10 events and stores up to 7 memories
Night	Player consolidates 3 memories into long-term storage
Travel	Player uses memory to survive hazards and progress
Endgame	Survive all 15 nights and reach the castle

4. Main Systems

System	Purpose
Short-Term Memory	Temporary daily memories
Long-Term Memory	Permanent and stronger memories
Memory Journal	Review and manage stored knowledge
Merge System	Combine related memories into stronger rules
Emotion System	Boosts vivid memories

System	Purpose
Repetition System	Improves memory reliability
Context Relations	Changes meaning based on conditions
Health System	Tracks survival
Delay System	Punishes poor planning and recall

5. Long-Term Memory Journal

The Long-Term Memory Journal is accessed through a small UI button near the memory panel. It acts as a living archive where the player can inspect, compare, and merge memories.

Journal Feature	Description
Memory List	Displays all long-term memories
Detail View	Shows traits, cues, and relationships
Merge Menu	Combines related memories
Search and Filters	Organize memories by type or danger
Knowledge Progress	Tracks how complete a memory is

6. Memory Trait System

Trait Type	Examples
Visual	Claws, smoke, red berries
Sound	Buzzing, howling, crackling
Smell	Smoke, sweetness, rot
Behavior	Dangerous when approached
Context	Safe near rivers, dangerous near caves
Risk Rule	Dangerous only when close
Reward Rule	Restores health or reveals paths

7. Memory Merge System

The merge system allows players to combine memories into more complete and intelligent rules. This turns simple labels into contextual understanding.

Merge Type	Example	Result
Exact Merge	Bear + Bear	Stronger and more reliable recall
Trait Merge	Smoke + Fire	Smoke warns of nearby fire
Context Merge	Bees + Honey	Bees dangerous when approached near nests
Progressive Merge	Repeated mushrooms	Clear distinction between safe and dangerous

8. Risky Merge System

Merging memories too early can create incorrect assumptions. This simulates false memory formation and overgeneralization.

Risky Merge	Possible Wrong Result
Safe mushroom + dangerous mushroom	All mushrooms are assumed safe
Friendly dog + wolf	All animals assumed harmless
Warm fire + wildfire	All fire assumed useful

9. Relation-Based Hazard Table

Object	Danger Condition	Safe / Helpful Condition
Fire	Dangerous when close	Helpful at distance or controlled
Bees	Dangerous when disturbed	Safe when ignored
Bear	Dangerous when approached	Safe when avoided early
Cliff	Dangerous when stepped near	Safe when recognized
Water	Dangerous when deep or infested	Helpful when calm
Wolf	More dangerous at night	Easier to avoid in daylight
Smoke	Signals nearby fire	Useful as an early warning

10. Memory Growth System

Exposure Count	Knowledge Level	Player Understanding
1	Unknown	Only vague understanding
2	Partial	Some traits are recognized
3	Reliable	Object can usually be identified correctly
4+	Mastered	Complex context rules understood

11. Progression by Night

Night	Focus	New Mechanic
1	Basic labels	Simple safe vs dangerous objects
2	Categorization	Food and hazard separation
3	Sensory cues	Sound and smell clues
4	Repetition	Repeated memory strengthening
5	Ambiguity	Uncertain mushrooms and berries

Night	Focus	New Mechanic
6	Emotion	Fear-based memories
7	Context	Objects behave differently by situation
8	Memory overload	Many similar events
9	Multi-cue recall	Combining clues
10	Delay penalties	Mistakes affect timing
11	False memory pressure	Misleading similarities
12	Advanced merging	Organizing memory webs
13	Fast recall	Limited decision time
14	Mixed systems	Complex layered encounters
15	Final mastery	All systems combined

12. Event Types

Event Type	Purpose
Simple Identification	Learn labels and danger types
Food Recognition	Identify safe resources
Hazard Recognition	Avoid environmental danger
Ambiguous Events	Learn through repeated exposure
Sound Cue Events	Recall danger from sound
Smell Cue Events	Predict danger using smell
Pattern Recognition	Connect clues and outcomes
Emotional Events	Strengthen emotional memory
Delayed Consequence Events	Plan using remembered information

13. Example Memory Archive Entries

Memory	Traits	Rule
Bear	Fur, claws, growl	Dangerous when approached
Fire	Smoke, heat, light	Dangerous when close
Bees	Buzzing, flowers	Dangerous when disturbed
Cliff	Loose rocks, steep edge	Dangerous when stepped near
Mushrooms	Shade, dampness	Requires repeated observation
Smoke	Smell, warning	Signals nearby fire

14. Optional Advanced Features

Feature	Purpose
Memory Notebook	Review all discovered memories
Dream Replay	Replay important memories during sleep
Hint System	Assist players after repeated failures
Difficulty Modes	Adjust challenge level
Memory Search	Find memories by cue or type
Hidden Mastery Ending	Reward high-level memory play
Journal Notes	Educational explanations after encounters
Late-Game Branches	Different paths based on memory quality

15. Win and Loss Conditions

Condition	Result
Reach the castle after 15 nights	Victory
Health reaches zero	Loss
High mastery and memory accuracy	Special ending

16. Final Design Philosophy

Limbic Journey is designed to teach that memory is not simply storing facts. The game emphasizes that memory is emotional, selective, contextual, and constantly changing. Players gradually move from memorizing labels to understanding relationships, patterns, and conditions.